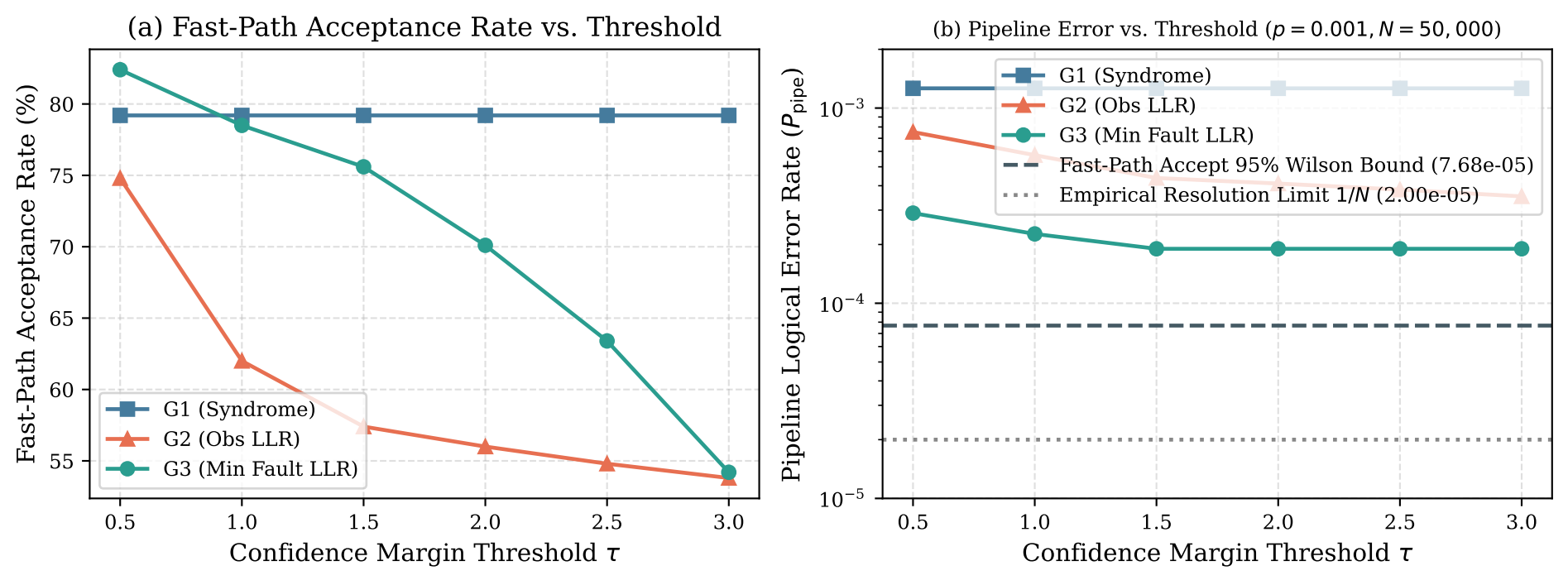
(a) Fast-Path Acceptance Rate vs. Threshold
Fast-Path Acceptance Rate (%)
80
75
70
65
60
55
0.5
1.0
1.5
2.0
2.5
3.0
Confidence Margin Threshold τ
G1 (Syndrome)
G2 (Obs LLR)
G3 (Min Fault LLR)
(b) Pipeline Error vs. Threshold ($p = 0.001, N = 50,000$)
Pipeline Logical Error Rate (P_{pipe})
10^{-3}
10^{-4}
10^{-5}
0.5
1.0
1.5
2.0
2.5
3.0
Confidence Margin Threshold τ
G1 (Syndrome)
G2 (Obs LLR)
G3 (Min Fault LLR)
Fast-Path Accept 95% Wilson Bound (7.68e-05)
Empirical Resolution Limit $1/N$ (2.00e-05)